
1. Project Planning & Management

Project Proposal

Project Title " Students Focus Monitoring System using Gaze Estimation"

Overview

This project aims to develop an AI-powered system that monitors students' attention during online sessions using gaze estimation and computer vision techniques. The system analyzes eye movement, head pose, and facial features to determine whether a student is focused or distracted.

It provides:

- Real-time monitoring of student attention
- Detection of suspicious behavior during exams (cheating)
- Attention analytics dashboard for teachers
- Intelligent feedback to improve lecture quality

The system will be implemented as both a **Web Application** and a **Desktop Application**, ensuring flexibility and usability across different environments.

Objectives

- Build a real-time student attention monitoring system
 - Develop multiple machine learning models and select the best-performing one
 - Optimize model performance through parameter tuning
 - Provide interpretable attention scores
 - Deliver a scalable and startup-ready solution
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Scope

In Scope

- Gaze estimation using camera input
 - Eye tracking + head pose estimation
 - Attention score calculation
 - Real-time inference system
 - Web dashboard for monitoring
 - Cheat detection during exams
 - Model comparison and optimization
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Out of Scope

- Hardware-based eye tracking devices
 - Cloud deployment (current phase)
 - Large-scale distributed systems
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Project Plan

Timeline (8 Weeks)

Week 1–2

- Dataset collection
 - Literature review (latest research papers)
 - Data understanding and preprocessing
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Week 3–4

- Feature extraction (eyes, face, head)
 - Build baseline models
 - Each team member develops a different model
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Week 5–6

- Model evaluation and comparison
 - Select best-performing model
 - Hyperparameter tuning
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Week 7

- Build Web App + Desktop App
 - Integrate model into system
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Week 8

- Testing and validation
 - Performance optimization
 - Final documentation
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Milestones

-  Data Collection Completed
 -  Models Developed (4 Models – one per member)
 -  Best Model Selected
 -  Hyperparameter Optimization Done
 -  System Implemented (Web + Desktop)
 -  Final Testing & Delivery
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Deliverables

- Trained ML models
- Best optimized model
- Web Application
- Desktop Application
- Dashboard with analytics

- Final report
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Task Assignment & Roles

Team Member Role

Hany	Model Development + Research + Integration
Yomna	Model Development + Data Processing
Mahfouz	Model Development + Feature Engineering
Yehia	Model Development + Evaluation

Work Strategy

- All team members:
 - Explore datasets
 - Study research papers
 - Build independent models
 - Then:
 - Compare models
 - Select best model
 - Perform parameter tuning collaboratively
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Risk Assessment & Mitigation Plan

Risk	Impact	Mitigation
Low model accuracy	High	Use multiple models + feature engineering
Poor lighting conditions	Medium	Apply normalization techniques

Risk	Impact	Mitigation
Occlusions (glasses, head movement)	Medium	Use robust face detection models
Real-time performance issues	High	Optimize model + reduce input size
Overfitting	Medium	Use cross-validation
Limited dataset diversity	High	Combine multiple datasets

KPIs (Key Performance Indicators)

Performance Metrics

- **Accuracy**
 - Target: $\geq 90\%$
 - Measures correctness of focus detection
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- **Response Time**
 - Target: $\leq 100\text{ ms}$
 - Ensures real-time monitoring capability
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System Metrics

- Real-time processing capability
 - Stability during long sessions
 - Smooth UI performance
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Model Metrics

- Precision / Recall
 - F1-score
 - False negative rate (missed distraction cases)
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Business / Product KPIs (Startup Ready)

- User engagement (attention tracking usage)
 - Adoption by teachers
 - Improvement in student focus over sessions
 - Reduction in cheating incidents
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Why This Project is Strong (Evaluator View)

- Uses **state-of-the-art AI (Gaze Estimation)**
- Includes **Model Comparison Strategy (very strong point)**
- Covers **End-to-End System (Model + App)**
- Has **Real-world impact (Education + Exams)**
- Designed as **Startup-ready solution**